Queensland University of Technology

EGH490-2 Research Project 2

Final Report

PV Shuffling Converter

Design, Simulation and Bench Validation of a Power Shuffler for Partially Shaded PV Strings

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October 2026



Abstract

This report outlines the design, simulation and bench validation of a non-isolated bidirectional power shuffler, a partial-power converter which transfers only the mismatch current between adjacent modules in a series photovoltaic (PV) string. In a series string, partial shading of one module limits the current of every module, and whilst bypass diodes protect the shaded cells, the output of the bypassed substring is lost. A datasheet-based PV model and three candidate topologies were simulated in Qspice, and an interleaved full-bridge buck–boost topology using a coupled inductor was selected for its low ripple current. The simulations predicted a peak recovery of approximately 545 W over bypass diodes on a 2.1 kW four-module string. A printed circuit board carrying two power shufflers for a three-module string was then designed, with its high-frequency decoupling and snubbers sized using a parasitic-aware board model, before being assembled and commissioned. In open-loop bench tests at 50 % duty cycle, three supply channels emulated a string in which the middle module was limited to 0.5 A against 1.3 A for the other two. At a fixed 70 V load, the string output was limited to 34 W by the weakest channel without the power shuffler, whereas with the power shuffler operating each channel delivered its full current and 68 W reached the load, 93.8 % of the 72.5 W supplied. Output peaked at 69 W at a 300 kHz switching frequency, whilst the spread between module voltages increased from 0.2 V at 200 kHz to 1.1 V at 400 kHz as the fixed dead time occupied a larger share of the switching period. The measured inductor ripple was within approximately 25 % of the coupled-inductor model. On three 80 W panels with one partially shaded, the string output at a fixed 60 V load increased from 40 W to 176 W in the best run and from 9 W to 112 W in another, whilst in a third run 93.6 % of the 106.4 W supplied by the panels reached the load. As fixed losses dominate at light load, adaptive switching frequency and dead time, hardware overcurrent protection and real-panel testing with a maximum power point tracker are recommended for future work.

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1 Introduction

Over the last decade, photovoltaic (PV) generation has grown rapidly, with annual installations increasing fifteen-fold on a global scale, from 40 GW in 2014 to 602 GW in 2024 [1]. In Australia, solar generation revenue grew from AUD 100 million to AUD 2.1 billion over the ten years from 2015 to 2025 [2]. Despite this growth, the electrical layout of PV arrays has not changed significantly: cells are connected in series within a module, modules are connected in series to form a string, and strings are connected in parallel [3]. In a series connection, the weakest element sets the current of the whole string. As such, a single shaded, soiled or degraded module reduces the output of every other module in its string.

When PV cells are shaded, they can be driven into reverse bias and form hotspots, which if not alleviated can cause localised damage to the module [4]. To combat this, bypass diodes are traditionally installed in antiparallel across a section of cells, restoring current flow and preventing hotspots. However, the output of the unshaded cells within that section is also discarded, leading to a large reduction in output power. In a typical 72-cell module divided into three sections of 24 cells, a localised shading event covering only one or two cells can activate a bypass diode and discard the output of the remaining cells in that section, reducing the module output by up to one third. In addition, conducting bypass diodes introduce multiple local maximum power points (MPPs) on the string power–voltage (P-V) curve, which degrades the performance of maximum power point tracking (MPPT) algorithms such as perturb and observe [5], [6].

To address this mismatch, a power shuffler may be used, which takes the form of a bidirectional DC–DC converter connected between adjacent modules. A power shuffler processes only the difference in current between the adjacent modules, allowing each module to operate near its own MPP whilst the remainder of the string power bypasses the converter [7], [8]. The concept is well established at a system level [8], [9], [10], but little published work describes the circuit design, component stresses and switching behaviour of a power shuffler sized for modern high-current modules. This project addresses that gap through simulation, printed circuit board (PCB) design and bench testing. Its key research question is:

> *How can a scalable, non-isolated bidirectional DC–DC power shuffler be designed, simulated and validated to mitigate partial shading losses in a PV string, and which topology and component choices best balance efficiency, cost and compatibility with commercially available modules?*

The aim of this project is to design, simulate and bench-validate a non-isolated bidirectional power shuffler which recovers power lost to partial shading in a series PV string. In addition, the project aims to characterise the circuit-level behaviour that existing power shuffler studies do not report. The objectives of this project are to:

1. develop and validate a datasheet-based model of a modern high-output PV module for circuit simulation;
2. compare candidate non-isolated bidirectional shuffler topologies and select one on inductor current ripple and switch stress;
3. quantify the power a shuffler recovers over bypass diodes, first with ideal switches and then with realistic devices and losses;
4. design and fabricate a PCB prototype of the selected topology for a three-module string; and

5. bench test the prototype to validate its switching behaviour and power shuffling against simulation.

The project was split into two linked streams, namely power electronics, which is covered in this report, and control systems, which was led by a second student (Jared Saunders). As such, this report covers the power shuffler design, simulation, hardware and testing, with control algorithms and string-level MPPT considered out of scope except where they set hardware requirements. The MPPT boost converter listed in the project proposal was removed from scope during the second semester, so an electronic load was used in place of the MPPT input during testing.

2 Literature Review

This section outlines the literature reviewed for this project. The first part covers the effect of partial shading on series-connected PV strings and the strategies used to mitigate it. Power shuffler topologies are then discussed, along with the PV module models used for simulation. Lastly, the research gap which this project aims to address is identified.

2.1 Partial shading in series-connected strings

A PV cell is commonly described using the single-diode model, which consists of a photocurrent source proportional to irradiance, shunted by a diode and a parallel resistance, and connected through a series resistance [4]. Lower irradiance reduces the photocurrent almost linearly, whilst the open-circuit voltage only falls logarithmically. When a shaded module is in series with unshaded modules, either the string current falls to the current of the shaded module, or the string forces more current through the shaded cells than they can generate. In the second case, these cells are driven into reverse bias, dissipating power and forming hotspots [4], [11]. To prevent this, commercial modules use bypass diodes across each substring of 20 to 24 cells [3]. Whilst this protection is effective, it is costly in energy, as a conducting bypass diode removes its entire substring from the string.

Partial shading also degrades MPPT performance. Under uniform irradiance, the string power–voltage curve has a single maximum, which could be reliably tracked by simple algorithms such as perturb and observe (P&O) [5], [12]. On the other hand, when bypass diodes are conducting, the curve has a local maximum for each distinct irradiance level, so P&O may settle on a local maximum rather than the global maximum [5], [6]. Whilst global-search algorithms exist, they add computation and cost without removing the current mismatch [6].

2.2 Mitigation strategies

Mitigation strategies can be grouped into three families, namely array reconfiguration, advanced MPPT control and module-level power electronics [7], [10]. Static and dynamic reconfiguration rearrange the connections between modules so that shading is evenly spread across strings or rows [11], [13]. Whilst dynamic reconfiguration can approach the unshaded yield, its adoption has been limited by the required switch matrix and wiring [10]. Advanced MPPT, on the other hand, only treats the symptom, as it finds the global maximum of a mismatched string but cannot recover the energy discarded by the bypass diodes. In comparison, module-level converters address the cause by allowing every module to operate at its own MPP [10], [14].

2.3 Module-level converter architectures

Kasper et al. classify panel-integrated converters by the share of power they process [7]. Full-power converters, such as series-connected buck, boost or buck–boost optimisers, process the entire module output. Whilst this gives full control of each module, conversion losses are incurred even when there is no mismatch [7], [10]. In comparison, partial-power converters only process the mismatch, so their losses approach zero under uniform irradiance [7], [9]. Series-connected partial-power converters inject a correcting voltage in series with the module and usually require an isolation transformer [15]. On the other hand, parallel-connected partial-power converters, also known as differential power processing converters or power shufflers, equalise the voltages of adjacent modules and pass the difference in their currents between them [7], [9].

Walker et al. proposed this power shuffler concept for PV strings, using non-isolated bidirectional buck–boost converters between neighbouring modules [8]. In this concept, the converter rating is set by the expected mismatch rather than the module rating, most of the power never passes through the converter, and the structure scales with string length. Bidirectional buck–boost, Ćuk and flyback converters are the most common realisations [7], [8], with the non-isolated options avoiding a transformer and having the lowest component count. Khan et al. also showed that gallium-nitride devices increase the efficiency of submodule-level converters at high switching frequencies [14]. This indicates that the practical gain is decided not only by the architecture, but also by the switching behaviour and device choice.

2.4 PV modelling for converter simulation

A converter simulation requires a PV source with a realistic I-V characteristic, as the operating point of the converter depends on it. Whilst the single-diode model is accurate, its implicit equation and its parameters, which are not given in datasheets, make modelling difficult and impractical [4]. Karmalkar and Haneefa proposed an explicit current–voltage model where the parameters could be obtained directly from the datasheet short-circuit, open-circuit and MPP values [16]. This model exchanges some physical details in exchange for easy parameterisation and fast evaluation, which suits switched-converter simulations which has high time steps counts.

2.5 Research gap

The literature establishes the losses of partial shading and demonstrates that parallel-connected partial-power converters can recover much of the mismatch loss [8], [9], [10]. However, most studies report system-level energy gains or idealised converter behaviour, with three aspects receiving little attention, namely: the choice of topology for modern modules with current often exceeding 13 A; the circuit-level behaviour of a hardware power shuffler, including switching transitions, dead time and parasitic ringing; and a direct comparison between simulation and a built prototype. This project aims to address these gaps by selecting a topology based on ripple and stress criteria, modelling it at a device level, and building and testing a PCB prototype.

3 Methodology

3.1 Approach

This project utilised a simulation-based design approach. In the first semester, a validated PV model and three converter models of increasing fidelity were developed to select a topology and estimate the recoverable power. These results were then used to set the minimum requirements of the PCB and to inform the bench testing performed in the second semester. Qspice was selected for circuit simulation as it is free, is easy to learn for existing LTspice users, has a robust solver developed for power electronic applications, is backward compatible with many LTspice models and, most critically, supports C++ blocks in which control algorithms could be developed by the control systems student. KiCad was used for PCB design for its open-source nature and its extensive library of existing footprints.

3.2 PV module model

The Aiko A540-MCE60Mw module was chosen as the reference module as it is among the highest-output modules on the consumer market, which sets a worst-case operating condition for the power shuffler. The datasheet I–V curves were digitised in MATLAB, and linear regression was performed to obtain the Karmalkar–Haneefa parameters used to model the module in Qspice [16]. The model was verified against the datasheet I–V and P–V curves at 1000, 800, 600, 400 and 200 W/m^2 irradiance, which showed close agreement across the operating voltage range.

3.3 Topology comparison

Three non-isolated bidirectional candidates from the literature were simulated between two modules under identical conditions, namely a half-bridge buck–boost, a Ćuk converter, and an interleaved buck–boost design, consisting of two half-bridge legs switched 180° apart [7], [8]. Inductor current ripple was used as the selection criterion, as it sets the inductor and capacitor sizes as well as the RMS current in the switches.

3.4 Ideal and semi-realistic shuffler models

The selected topology was then modelled in a four-module string at two levels of detail. The ideal model used zero-resistance switches and ideal gate drives, giving an upper bound on the recoverable power. The semi-realistic model used MOSFET models generated from the manufacturer's datasheet, capturing details such as gate charge, on-resistance and body diode dynamics, combined with a behavioural gate driver with finite transition times and dead time. Where possible, parasitic elements such as series resistance were included for each component from its datasheet; where this was not possible, estimates were used to best approximate the power shuffler's dynamics.

Each model was then compared with a conventional PV string equipped only with bypass diodes. A shading event was simulated in Qspice in which the irradiance of one module gradually reduces to 200 W/m^2, and the recovered power was taken as the difference in output power between the two strings. The main modelling assumptions are summarised in Table 1 below.

Table 1: Main assumptions used in shuffler models simulations

Assumption	Justification	Effect on results
One module shaded, with a slow, gradual irradiance change.	Represents slow-moving shade from fixed objects, the most common residential case.	The recovery figure applies to this profile; fast transients and multi-module shading were not assessed.
PCB parasitics, thermal effects and inductor saturation excluded.	The layout did not yet exist, and parasitic elements slow the simulation sharply.	Losses are underestimated and switching-node ringing is not predicted. Section 3.6 adds them in a board-level model.
Open-loop operation at 50 % duty cycle.	Duty control belongs to the controls stream, and 50 % equalises adjacent module voltages.	No MPPT or closed-loop dynamics are represented.
String loaded by an ideal current sink.	Stands in for the MPPT converter input, which was out of scope.	String operating point is imposed rather than tracked.

3.5 PCB design

The board was rated for the Aiko A540-MCE60Mw module as a worst case basis, as it is one of the highest-output modules available, even though all testing was planned at lower voltages. At standard test conditions (STC), this module has an open-circuit voltage V_{OC} of 45.6 V and a short-circuit current I_{SC} of 14.84 A. Designing to these limits also leaves a margin for overvoltage and overcurrent events which may be caused by assembly and testing errors. From this rating and the project scope, the following five requirements were identified:

- **Switch voltage.** In all candidate topologies, the switches form half-bridges which span two modules. As the switch node swings between the two outer rails every cycle, each MOSFET must block the voltage of two modules, which is up to 91.2 V at STC.
- **Switch current.** A drain current rating of at least 25 A, approximately 1.7 times the module short-circuit current, to ensure that current spikes from wiring or assembly faults do not destroy the switches.
- **Efficiency.** Low on-resistance $R_{DS(on)}$ and low gate charge, to reduce conduction and switching losses and allow for higher switching frequencies.
- **Safe testing.** All testing to be performed below 120 V DC, the limit above which a licensed electrician is required [17].
- **Testability.** Control hardware which can be replaced quickly and a clear separation between power and logic circuits, as a first board revision is likely to experience faults during testing.

Inductor and capacitor values were obtained from the first-semester models (Sections 3.3 and 3.4), which related ripple current and switching frequency to component size. Components were then selected with headroom above these minimum values, for reliability and to absorb later design changes. In addition, reviewing these models with the supervisor exposed the ripple-current risk which led to the selection of the interleaved buck–boost topology (Section 4.1).

The number of shufflers on one board was decided based on three criteria, namely: the resulting string voltage against the 120 V DC limit imposed by AS3000 [17]; the complexity and cost of a first revision, where a single fault can disable the entire board; and the number of PWM timers and pins on the STM32 microcontrollers preferred by the controls stream. The architectures

considered against these criteria and the final board are discussed in Section 4.4.

Power components, namely the input connectors, MOSFETs, inductors and bulk capacitors, were placed well away from the digital circuitry, such as the voltage sensors and the microcontroller board. Components which must bridge the two domains, such as the isolated gate drivers and digital isolators, were placed along an isolation boundary running down the board, so that the power and logic sides could be clearly identified. Where possible, high-voltage and digital signals were kept on different layers. ADC and digital communication traces were routed on the bottom layer, away from the switching loops on the top layer, and change layers through vias placed as close to their ICs as possible. This is to limit noise coupling between the power and logic circuits.

Component placement was performed by a student experienced in PCB design, before the traces and copper zones were routed by the power electronics student, with input from the controls student on the controller interface. The layout was then reviewed by the supervisor and the experienced student. Any disagreements were settled through a discussion of both views before a decision was made, and any changes a reviewer made directly to the layout or components were communicated back to ensure that all parties stayed informed.

3.6 Parasitic-aware board model

After the first board was sent for manufacturing, a board-level Qspice model was developed to determine the values for the high-frequency (HF) decoupling capacitors and RC snubbers, as their footprints had been left as placeholders. This model reproduced the power stage along with its parasitic elements, which were estimated from the layout and datasheets as follows:

- Approximately 1.5 nH of loop inductance in each MOSFET drain path and 3 nH in each source path, 2.4 nH between the bulk capacitors, and 15 nH of equivalent series inductance (ESL) in each bulk capacitor;
- 0.85 nH of ESL and 13 mΩ of series resistance for the ceramic capacitors, and 750 nH with 6.95 mΩ for each module lead;
- A behavioural model of the 1ED3140 gate driver with separate turn-on and turn-off resistors (10 Ω and 4.7 Ω), 12 V drive, with approximately 70 ns propagation delay and the driver's cross-conduction interlock;

To ensure that the Qspice solver converged, each parasitic inductor was given a parallel damping resistor of 5 Ω on the board and 50 Ω on each of the module input cables. It is noted that the parallel resistance provides only slight damping of the resonances, so the simulated ringing is greater than that observed during bench testing. The model was then run at two conditions using the reference module, namely balanced irradiance (no shading) and one module operating at 200 W/m^2. In both conditions, the MOSFETs were switched at a 200 kHz switching frequency, 50 % duty cycle and 200 ns dead time, as these represent the standard switching arrangement used on the physical PCB for bench and module testing. Under balanced irradiance, zero-voltage switching (ZVS) was achieved, whereas in the shaded scenario hard switching occurred on one edge of one leg. This hard-switched edge was used to trial candidate resistor and capacitor values for the RC snubbers, whilst the HF decoupling capacitors were determined by inspection of the rail ringing and switch-node overshoot. The chosen RC snubber and HF decoupling values are outlined in Section 4.5.

3.7 Assembly and bench testing

Boards were assembled using a stencil and solder paste. The board and stencil were first fixed in position using spare boards taped around them as a jig, as seen in Figure 1(a), and cleaned with isopropyl alcohol. Solder paste was then spread through the stencil and the components were placed, before the board was reflowed in the oven shown in Figure 1(b), inspected and reworked where required.

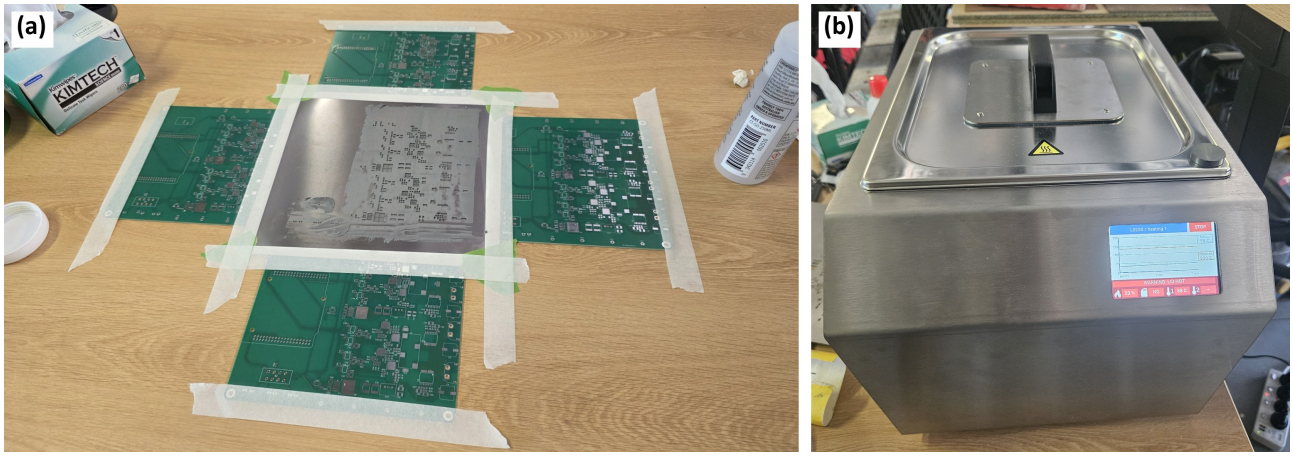


Figure 1: Assembly of power shuffler: (a) PCB assembly with the stencil and improvised jig; (b) oven used to reflow the board.

Testing was performed incrementally. Each power shuffler was first run on its own with no load at a fixed 50 % duty cycle, and its switching waveforms were inspected on an oscilloscope to confirm correct operation. The board was then tested as a string, with each module emulated by one channel of a programmable three-channel power supply. The channels were connected in series at equal voltage but with different current limits, which mimics a string with partially shaded modules, as shading mainly lowers the current of a module rather than its voltage. Without the power shuffler, the channel with the lowest current limit sets the string current, similar to a shaded module in a real string. With the power shuffler operating, each channel was able to deliver power up to its current limit, at which point it entered constant-current operation.

The string output was connected to an electronic load, and the output power was compared with the power shuffler off and on. The switching frequency was varied between 100 kHz and 400 kHz, with a fixed 200 ns deadtime. Waveforms were captured on an oscilloscope, whilst operating data were logged by the controls stream through the microcontroller and a Raspberry Pi to a Grafana dashboard. The equipment used was an R&S NGE103B three-channel supply, a Rigol DL3021 electronic load and an R&S RTB2004 oscilloscope. Real-panel tests were performed using three 80 W panels, with the load set to a fixed 60 V (Section 4.7).

3.8 Risk, ethics, sustainability and stakeholder engagement

As the first semester was simulation only, experimental risk was negligible. Hardware testing, on the other hand, introduced electrical and thermal hazards, which were managed using an incremental protocol. No-load tests were performed first, followed by low-power bench tests with current-limited supplies emulating the PV modules at voltages below the 120 V DC extra-low-voltage limit [17], before tests with real 80 W panels. All testing was planned below this

limit, and any future testing above it, such as with full-size modules, must be supervised by a licensed electrician [18]. Two project risks were active in the first semester. The control stream's dependency on the power electronics stream materialised when the topology selection remained open for several weeks because of the ripple-current problem. As a topology change alters the system dynamics on which the control design depends, this risk could not be contained; the topology was settled close to the end of the semester, with the power electronics student assisting with the foundational development of the control system.

A design error in the power electronics stream was also identified, in which the digital isolator for the floating voltage sensor was wired incorrectly. This was addressed by removing the isolator IC and connecting the voltage sensor directly; however, the top voltage sensor remained non-functional, and the top module voltage was instead derived from the electronic load readings.

Lastly, testing errors also occurred, including short circuits, reversed input polarity and incorrect inductor wiring, which subjected components to stress. These were mostly contained by the generous component ratings chosen during design, the fault protection coded into the microcontroller and the safety switches for rapid shutdown. Despite this, one development board was damaged beyond repair when a short occurred on a digital communication line. This was resolved as the controls student had purchased several development boards to safeguard against this risk.

Ethical decisions followed the Engineers Australia Code of Ethics [19]. Results are reported along with their limitations, and external work, such as the Karmalkar–Haneefa model and the power shuffler architecture of Walker et al., is attributed, with the project's own contribution stated separately.

Sustainability was guided by Engineers Australia's Implementing Sustainability [20]. The main benefit of this project is a higher yield from existing installations, which reduces the panel area, embodied energy and material needed for the same output. In addition, industrial and automotive-grade MOSFETs, gate drivers and capacitors were preferred for their service life, parts were reused across board revisions, and the hardware was sized for the highest-rated module so that it remains compatible with lower-rated modules already installed.

The supervisor was the primary stakeholder, with fortnightly meetings used as design reviews at each milestone, whilst the controls stream was an active stakeholder at the hardware interface. In the second semester, the layout was reviewed an external student experienced in PCB design.

4 Results, Analysis and Discussion

4.1 Topology selection

As seen in Figure 2, under identical operating conditions, the two non-interleaved candidates carried inductor current peaks of approximately 15.5 A, compared to approximately 9.5 A in each inductor of the interleaved topology. As the two legs are switched 180° out of phase, their antiphase ripple currents cancel at the converter output at 50 % duty cycle. As such, the bulk capacitors experience far less ripple current than with a non-interleaved topology. For a non-interleaved topology to achieve the same output ripple current, a significantly larger inductor or a higher switching frequency would be required, adding complexity, losses and cost. Therefore, the interleaved buck–boost topology shown in Figure 3 was selected.

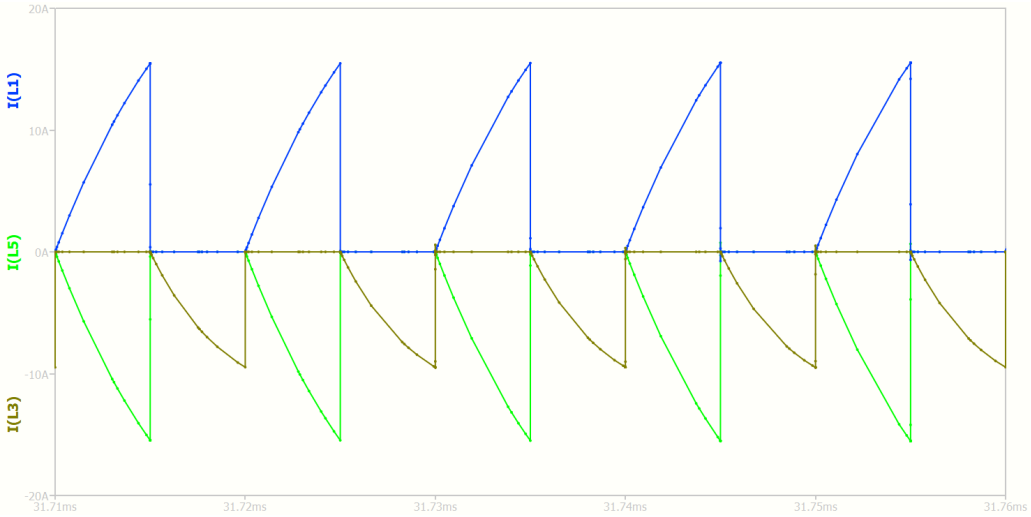


Figure 2: Inductor current under identical conditions: half-bridge buck–boost (blue), Ćuk converter (green), interleaved full-bridge buck–boost (brown, only one leg shown).

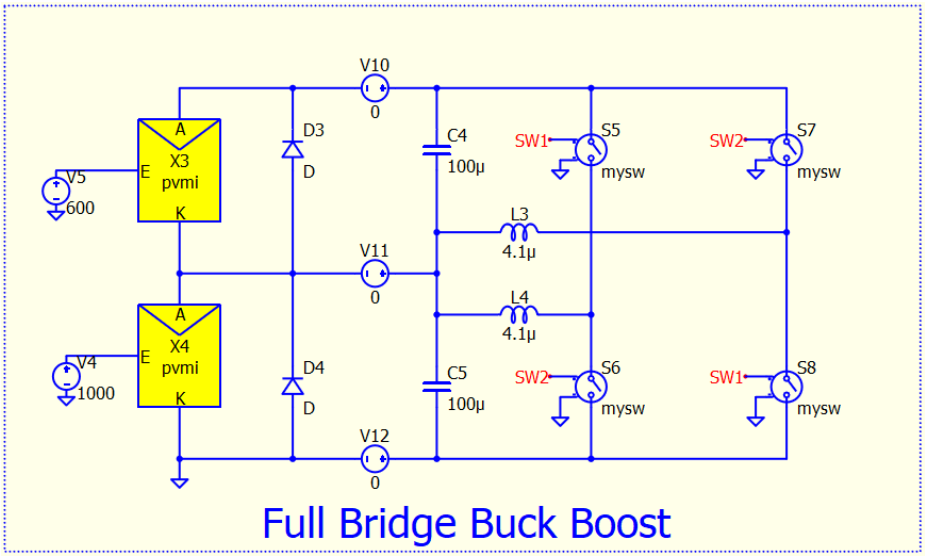


Figure 3: Selected interleaved buck–boost shuffler

4.2 Recoverable power

Figure 4 compares two four-module PV strings: one equipped with three power shufflers (top trace) and a reference string relying only on bypass diodes (middle trace). During the simulation, the irradiance of one module is reduced linearly from 1000 W/m^2 to a terminal value of 200 W/m^2. Prior to shading, both strings delivered approximately 2.1 kW, demonstrating the negligible loss of the power shuffler at uniform irradiance, a common advantage of partial-power processing. After 50 ms, the bypass diode in the reference string begins to conduct, diverting current around the shaded module, and the reference string settles at 1.47 kW once the diode is fully forward biased. In contrast, the shuffled string shows a more gradual reduction in output as the shading deepens, with a peak power recovery of approximately 545 W at 70 ms. The recovered power then falls close to zero as the shaded module reaches the terminal irradiance of 200 W/m^2, at which point the other modules cannot redistribute any further current.

As seen in Figure 5, the semi-realistic model only slightly reduced the peak recovery, to approximately 515 W. However, the loss penalty grew with the shuffled current. At the terminal

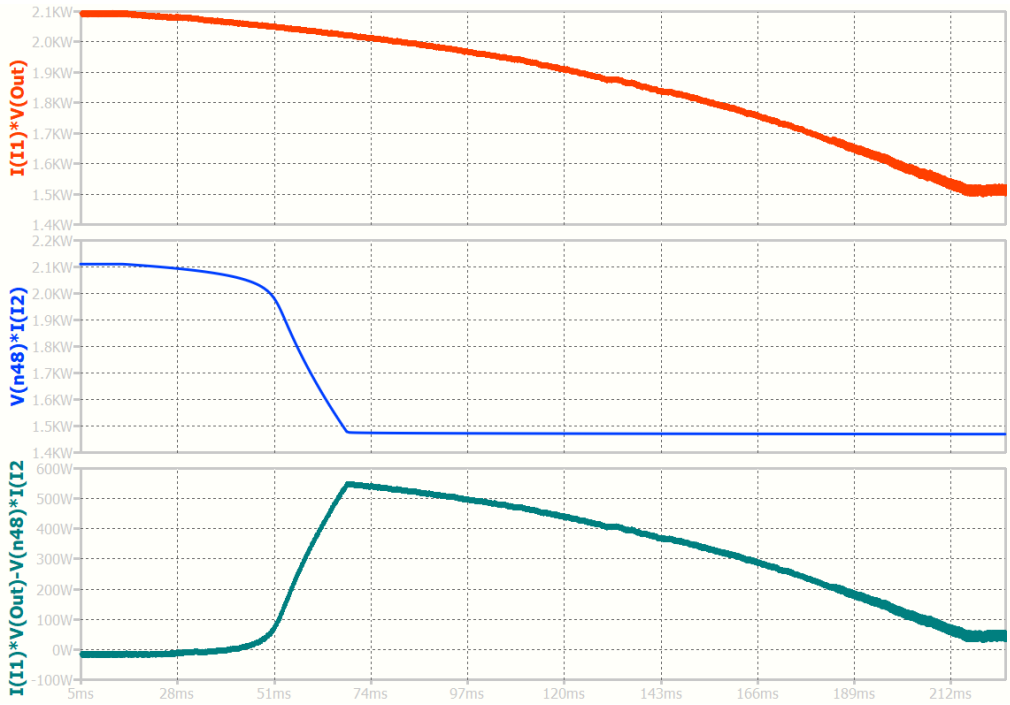


Figure 4: Ideal shuffler string (top) vs reference string (bottom) output; bottom: recovered power.

shading of 200 W/m^2, the semi-realistic power shuffler delivered approximately 60 W less than the bypass-diode string, whereas the ideal model still gained about 50 W. As such, under extreme mismatch the power shuffler should be disabled and the bypass diode allowed to conduct, reducing component stress.

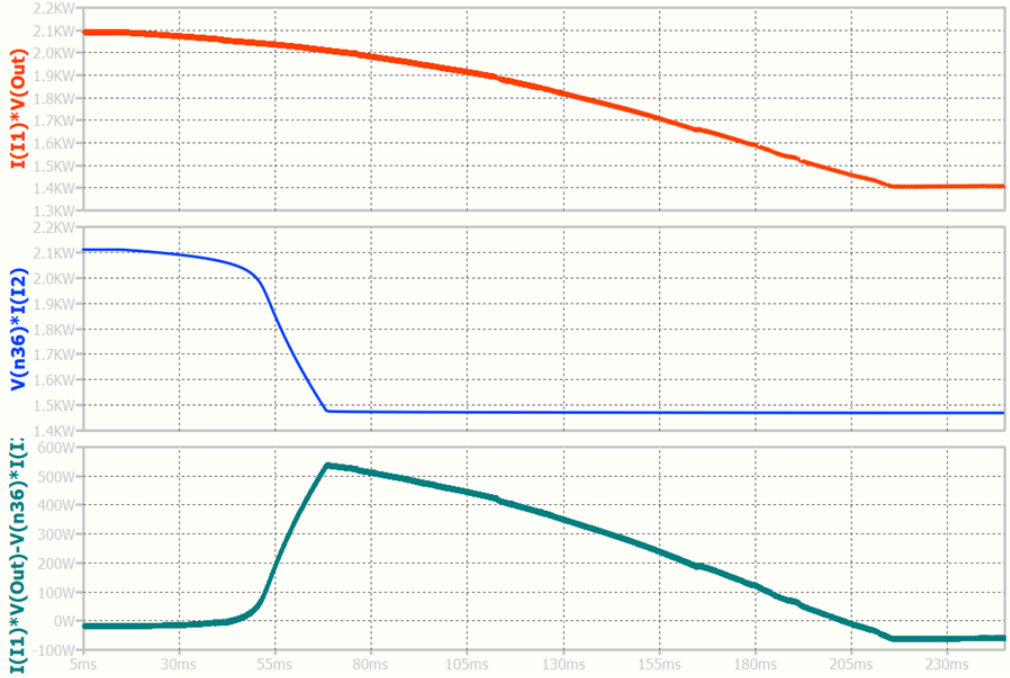


Figure 5: Ideal semi-realistic string (top) vs reference string (bottom) output; bottom: recovered power.

The semi-realistic model was also sensitive to dead time. A fixed dead time increased body-diode conduction at light shuffling currents, resulting in an additional loss of approximately

15 W, whilst being too short at heavy currents. The first-semester conclusion was therefore that the dead time should adapt to the operating point. The prototype currently uses a fixed 200 ns dead time, with adaptive dead time recommended for future work (Section 5).

4.3 Design changes carried into the hardware

Two decisions at the end of the first semester changed the basis of the hardware design. Firstly, the discrete 250 μH inductors were replaced by a 4.1 μH coupled inductor pair, the TDK ERUC23-4R1K, for a smaller footprint, lower winding loss and a higher current rating. Most critically, the coupled winding also provides differential current cancellation at the output, which greatly reduces the output RMS ripple current and widens the choice of bulk capacitors, which have finite ripple-current ratings. Secondly, the switching frequency was raised from 50 kHz to 200 kHz to keep the capacitor ripple manageable with the smaller inductance.

4.4 PCB design outcome

The first board architecture combined five PV inputs, four shufflers and an onboard microcontroller, as seen in Figure 6, to demonstrate current transfer along a longer string, including from one end to the other. However, this architecture was abandoned after the initial schematic for three reasons. Firstly, with the reference module, the five-module string reaches an open-circuit voltage of 228 V, which is almost twice the 120 V DC limit above which a licensed electrician is required [17], and even older modules with a V_{OC} of 30 to 40 V would exceed this limit. Secondly, a five-input board would be complex and costly for a first revision, where one fault could disable the entire board. Finally, the controls student found that most STM32 microcontrollers lack enough timers and pins to drive four shufflers.

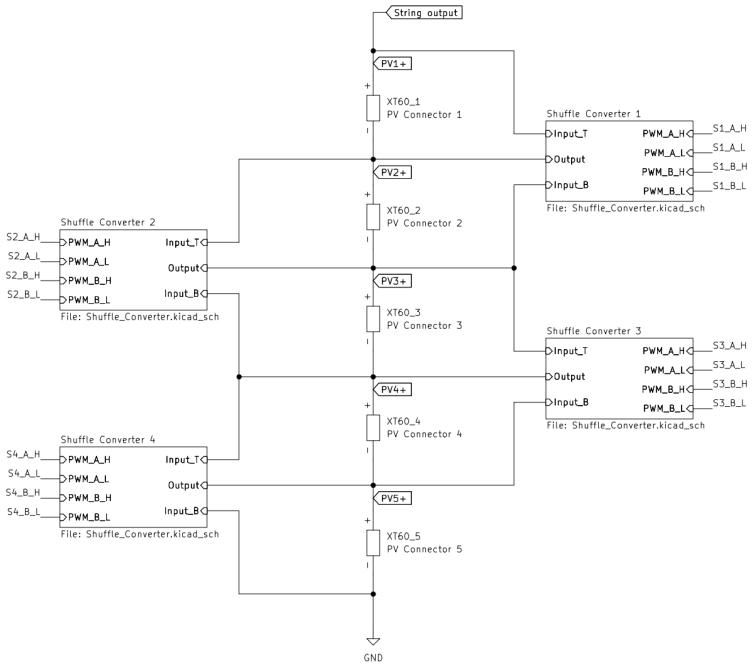


Figure 6: Initial PCB architecture.

The as-built PCB, as seen in Figure 7, uses two power shufflers serving three modules, which

is the smallest arrangement that can still demonstrate multi-shuffler control by the control stream. Stacking several boards via headers to extend the shuffler chain was also considered, which would have lowered the relative voltage within each board; however, this required high-current isolation between boards using relays and additional wiring, and so was not pursued. As the Aiko module served only as a worst-case rating, the three-module board could be tested entirely below the 120 V DC limit using the bench power supply (three channels of up to 32 V) and 80 W test panels ($V_{\mathrm{OC,string}} \approx 75\,\mathrm{V}$). In addition, the planned onboard microcontroller was replaced by a development board on headers, so that a controller damaged by a test fault could be swapped in minutes rather than reworked.

As the PV input voltages are too high to drive the MOSFET gate drivers directly, each power shuffler was provided with a wide-input buck converter supplying 12 V for the gate drivers and their bootstrap circuits. A fixed 3.3 V low-dropout (LDO) regulator downstream of each buck converter powers the digital ICs, such as the current sensors and the digital isolator. This is critical for shuffler 1, as most voltage sensor ICs have a maximum rating of 85 V, which is lower than the $3 \times V_{\mathrm{OC}}$ of most residential PV modules. As such, a floating 3.3 V supply is provided, which powers both the voltage sensor and its digital isolator.

In addition, two switches are installed, as seen in Figure 7, with the upper switch being an SPDT and the lower switch a DP3T. The DP3T switch allows the 3.3 V rail to be powered either from the Nucleo development board (the controller) or from the onboard LDO. This safeguards against a failing LDO causing loss of the 3.3 V rail, and also acts as an emergency shut-off, as the middle throw position is grounded, allowing all digital components to be shut down rapidly. Similarly, the SPDT switch controls the 3.3 V rail of the gate drivers, so that in an emergency the gate drivers can be turned off quickly.

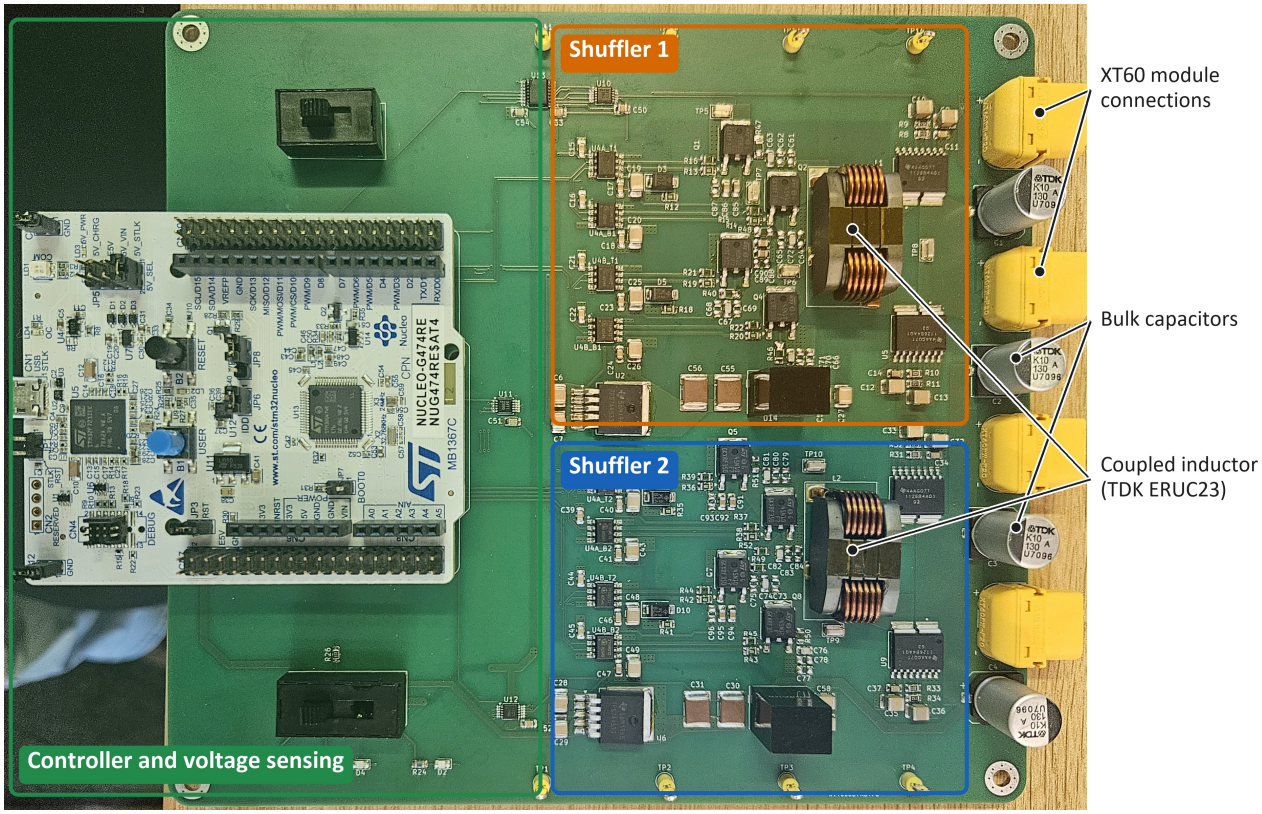


Figure 7: Assembled power shuffler PCB.

The key components are summarised in Table 2 below, and the full bill of materials is given in

Appendix 4.

Table 2: Key component and selection rationale

Function	Part	Key rating	Reason
MOSFET Switches	STD45N10F7	100 V, 45 A, DPAK	• Provides sufficient voltage rating to block two reference PV modules (91.2 V at STC). • low $R_{DS(on)}$ and gate charge
Coupled inductor	TDK ERUC23-4R1K	4.1 μH; saturation 22.5 A per winding	• Reduces ripple current due to phase cancellation • Smaller footprints than other inductor types
Gate drivers	Infineon 1ED3140MU12F	Isolated, separate source and sink outputs	• Isolated, high performance gate drivers • Interlock logic, preventing shorts
Current sensing	TI TMCS1126B4	Isolated Hall-effect sensor, 100 mV/A	• Isolation between "hot" and logic components • No shunt resistors
Bulk capacitors	TDK B40910 A0137M000	130 μF, 80 V, hybrid polymer	• Low ESR • High ripple current rating • High voltage rating
Controller	NUCLEO-G474RE	STM32G474 development board	Replaceable after test faults; PWM timers for both shufflers
Buck Converter	Wuerth 173951275	18 – 74.5 V input	• Provides voltage source for boot strapping • Wide input voltage range • High current output (4 A), provides additional headroom for additional ICs/loads in the future.

4.5 HF decoupling and snubber sizing

Without HF decoupling capacitors, the parasitic-aware Qspice model rang at approximately 56 MHz, with the bulk-capacitor ESL and switch-leg inductance resonating against the MOSFET output capacitance. With the ceramic decoupling capacitors fitted, two resonances remained: a fast ring of the switching loop at approximately 57 MHz in the hard-switched case, and a slower ring at approximately 1.9 MHz between the ceramic decoupling bank and the bulk-capacitor ESL, with a quality factor of about 5. The candidate measures are compared in Table 3 below.

The results show that the two problems respond to different measures. As the rail ring is set by the HF ceramic bank against the bulk-capacitor ESL, it can be reduced with more ceramic capacitance or a resistive damper (a rail RC snubber), whilst the MOSFET snubbers and gate resistors do not reduce it. On the other hand, the hard-switched overshoot follows the turn-on edge rate, so a larger turn-on resistor roughly halves it for approximately 0.4 W of additional loss, whereas additional snubber capacitance helps less. Slowing the turn-off edge was found only to add loss. With the corrected MOSFET model built from the manufacturer's datasheet, the rail RC snubbers were not needed. As such, one 470 pF capacitor and one 4.7 Ω resistor,

Table 3: Qspice results for HF decoupling, snubber and gate-drive options

Configuration	Rail ring, balanced (pk-pk)	Overshoot above voltage rail	MOSFET loss
• 2x 100nf HF decoupling capacitors per converter • 1x 470 pF + 4.7 Ω RC snubber across each mosfet • 4.7 Ω turn-off resistor • (Installed on PCB)	3.6 V	15.2 V	4.1 W
1x additional 100 nF HF decoupling capacitor	1.8 V	14.5 V	4.1 W
Voltage rail RC snubber, 1uF + 0.33 Ω	1.5 V	14.7 V	4.1 W
Additional 470 pF snubber capacitor per MOSFET	3.4 V	10.4 V	4.4 W
Turn-on resistor raised from 10 Ω to 22 Ω	3.6 V	6.9 V	4.5 W
Turn-off resistor raised from 4.7 Ω to 22 Ω	3.6 V	16.0 V	6.3 W

forming an RC snubber across each MOSFET, were specified, with the 22 Ω turn-on resistor recommended for hard-switched operation in a future revision.

During bench testing, with the 250 V X7T ceramics fitted, the rail ring appeared at approximately 3.2 MHz rather than the predicted 1.9 MHz. As the ring frequency depends on the product of the loop inductance and the ceramic capacitance, and the X7T parts lose little capacitance at the tested voltages, this difference suggests that the model overestimated the loop inductance. This may be due to the bulk-capacitor ESL and trace inductance being estimated rather than measured. It is noted that the 22 Ω turn-on resistor was not fitted, as no hard switching was observed in the bench tests.

4.6 Open-loop bench results

The bench tests ran both power shufflers open loop at 50 % duty cycle with a 200 ns dead time. Three supply channels at 25 V were used to emulate the modules, with current limits of 1.3 A on channels 1 and 3 and 0.5 A on channel 2 to mimic a shaded middle module, whilst the electronic load held the string at 70 V. The switching waveforms at switching frequencies between 100 kHz and 400 kHz could be seen in Figure 8. In (b) and (c), the two switch nodes swing rail to rail 180 ° apart, and the two phase currents form antiphase triangles, as intended by the interleaved coupled-inductor design.

The winding ripple measured by the Hall-effect sensors is compared with the coupled-inductor model in Figure 8(d), where the model predicts a peak-to-peak winding current of $V_{\text{rail}}/(4L_{\text{eff}}f)$ with $L_{\text{eff}} = 13.75\,\mu\text{H}$. The measured ripple was within approximately 25 % of the model at all three switching frequencies, with the difference likely due to switching noise at the triangle peaks inflating the oscilloscope reading. This confirms the inverse coupling and its ripple behaviour.

The power results are summarised in Figure 9. Without the power shuffler, the 0.5 A channel 2 set the string current, and the string delivered approximately 34 W at 70 V. With the power

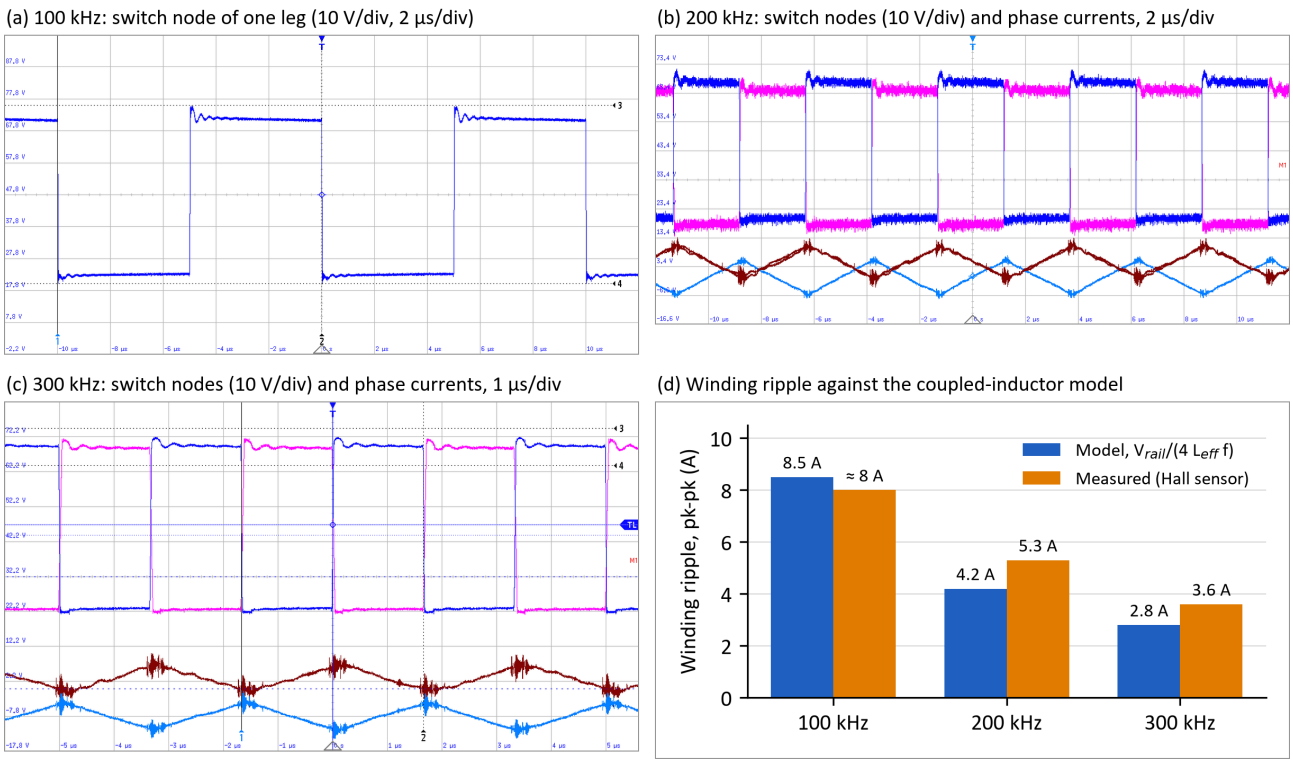


Figure 8: Open-loop bench waveforms at 50 % duty cycle: (a) switch node of one leg at 100 kHz; (b) and (c) switch nodes of both legs and Hall-effect sensor outputs, at 200 kHz and 300 kHz; (d) measured winding ripple against the coupled-inductor model.

shuffler operating, each channel delivered its own current limit and 67.9 W reached the load from the 72.5 W supplied, a board loss of 4.5 W (93.8 %). This is the intended function of the power shuffler: the string is no longer limited by the weakest channel, and each channel operates at its own maximum output. Figure 9(b) shows that the output was highest at a 300 kHz switching frequency, whilst the spread between module voltages grew with switching frequency as the fixed 200 ns dead time occupies a larger share of the switching period (4 % at 200 kHz, 8 % at 400 kHz). At 100 kHz, the board loss was approximately 8 W and the inductors were noticeably hotter than at 200 kHz, consistent with lower ripple reducing core loss.

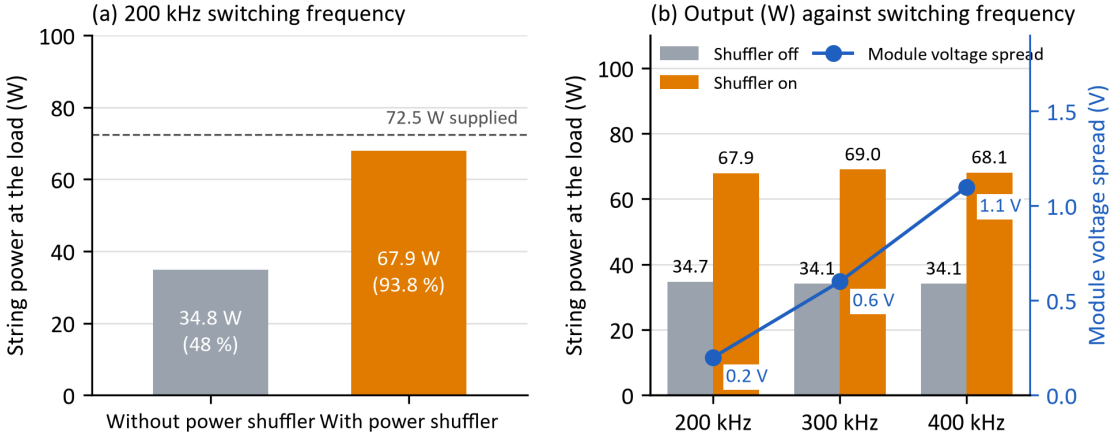


Figure 9: Open-loop bench results: (a) power delivered at a 200 kHz switching frequency; (b) output at 70 V and module voltage spread against switching frequency.

Setting one converter to a 51 % duty cycle moved its two module voltages apart by approximately 1.0 V, against the 0.96 V predicted by a 51:49 ratio on a 47 V rail. This confirms that the converter imposes the duty ratio on the module voltages and shows the sensitivity of open-loop operation to duty cycle errors.

As the converters only processed approximately 15 W at a 200 kHz switching frequency, the 4.5 W board loss corresponds to an efficiency of approximately 70 % on the processed power, even though 93.8 % of the supplied power reached the load. This is due to fixed losses from circulating ripple current, gate drive and dead-time conduction dominating at light load. In the early 200 kHz runs, the telemetry also showed an apparent DC current of up to 3.3 A in one leg of converter 2, against an expected 0.3 A per winding. This was caused by the ADC gain being set for a 100 mV/A sensor, whereas the Hall-effect sensor fitted outputs 25 mV/A, amplifying the reading by a factor of four. As seen in Figure 10, dividing the recorded current by four brings the average current of that leg back in line with the other legs. After gain correction, the 300 kHz telemetry showed the legs sharing the shuffled current with no circulating component.

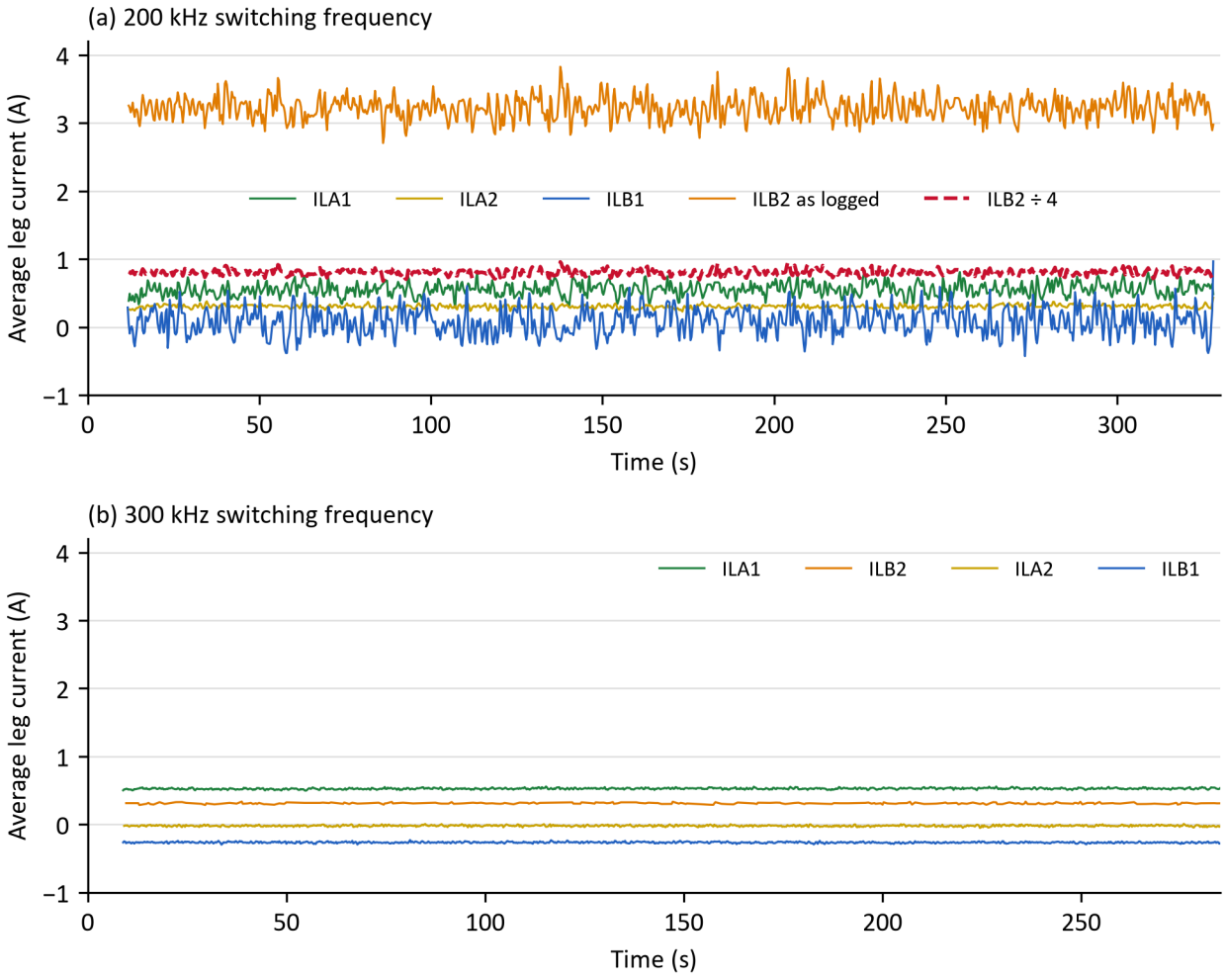


Figure 10: Average leg currents from the telemetry at (a) 200 kHz and (b) 300 kHz switching frequency.

The overshoot at each switching edge, measured from Figure 8, is summarised in Table 4, where the percentage overshoot (%OS) is relative to the rail-to-rail swing. The %OS falls from approximately 8 % at 100 kHz to 5 % at 300 kHz, as the ripple current which charges the switch node during each transition reduces with switching frequency. The peak drain-source voltage

of approximately 53 V leaves almost 50 % margin against the 100 V MOSFET rating. The rail ringing behaved as discussed in Section 4.5, at 2.6 V peak to peak and approximately 3.3 MHz at a module rail, and under 3 % of the 70 V string output.

Table 4: Switching characteristics measurement from Figure 8

Frequency	Switch-node swing	Overshoot (%OS)	Undershoot	String output ring (pk-pk)
100 kHz	49 V	4.0 V (8 %)	2.5 V (5 %)	1.5 V
200 kHz	49 V	2.9 V (6 %)	2.0 V (4 %)	1.6 V
300 kHz	47 V	2.5 V (5 %)	1.4 V (3 %)	1.7 V
400 kHz	47 V	3.0 V (6 %)	–	2.0 V

The switch-node overshoot measurement at 400 kHz was 3.0 V, and whilst no oscilloscope capture was saved, the bench results indicate that a 400 kHz switching frequency is the practical upper limit for the MOSFETs as driven on this board. The transitions themselves take under approximately 30 ns (Figure 8(c)); however, the two fixed 200 ns dead times occupy 16 % of the 2.5 μs period at 400 kHz, compared to 8 % at 200 kHz, which distorts the effective duty cycle and is reflected in the 1.1 V module voltage spread and lower output (Figure 9(b)). The gate-drive loss also doubles, from approximately 0.7 W at 200 kHz to 1.4 W at 400 kHz. In addition, as seen in Figure 11, each pair of simultaneous edges excites a ring on the string output which takes approximately 1 μs to settle, whereas the edge pairs arrive every 1.25 μs at 400 kHz. As such, operating above 400 kHz would require a shorter, adaptive dead time, lower gate-charge MOSFETs and additional damping of the output ring (Section 4.5).

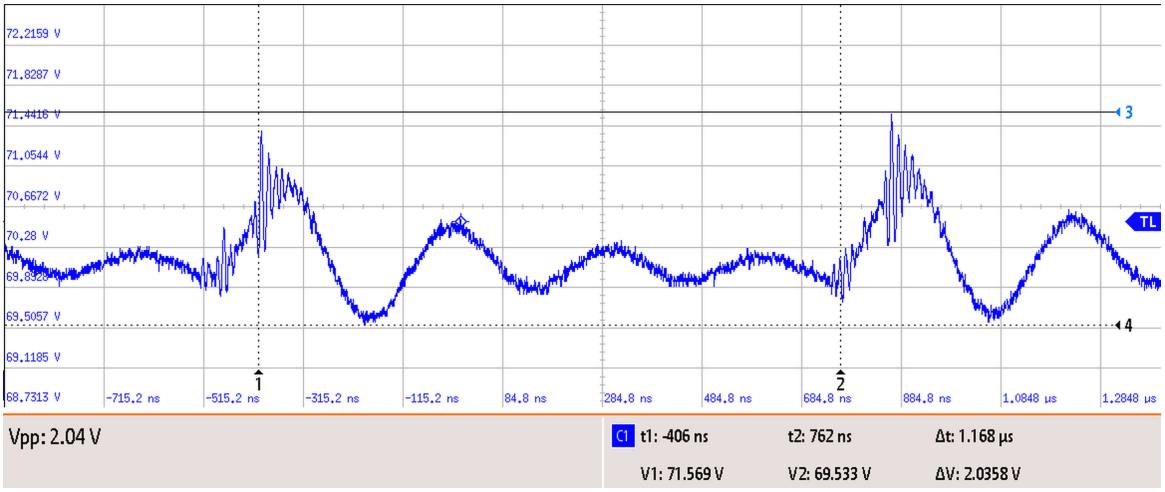


Figure 11: Ringing on the 70 V string output at 400 kHz, with the cursors 1.17 μs apart on successive edge pairs (2 V/div, 200 ns/div).

4.7 PV module test

A single outdoor session, cut short by the weather, ran the board open loop at a 200 kHz switching frequency on three 80 W KickAss PV panels (part number KAUSSP80; see Appendix 3), using the set-up shown in Figure 12. As an MPPT was not available, and the string exceeded the 200 W rating of the electronic load in the brightest periods, the electronic load held the

string away from the MPP at a fixed 60 V throughout, at which the string produced approximately 195 W under uniform irradiance in full sun. Panel 1, the panel at the highest potential, was shaded using a cloth placed over one corner.



Figure 12: Outdoor test set-up

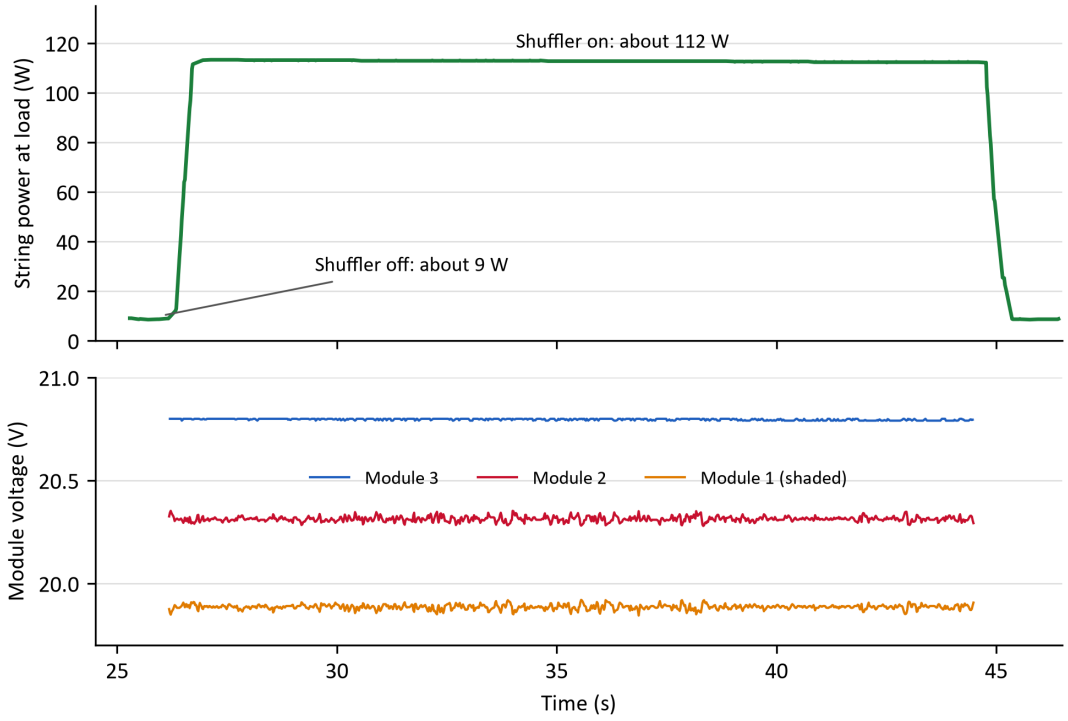


Figure 13: PV Module test, Run 2.

The three runs are compared in Figure 14. In run 2, shown in Figure 13, the shaded panel limited the string to approximately 9 W with the power shuffler off, whereas with the power shuffler operating the string delivered approximately 112 W with the module voltages held at 19.9 V, 20.3 V and 20.8 V. In run 3, the panel output of 106.4 W was calculated from the string current, measured with a Hall-effect current probe on the oscilloscope, and the module voltage sensor readings; 99.6 W of this reached the load, an efficiency of approximately 93.6 %, close to the 93.8 % measured on the bench. In run 1, the string output rose from approximately 40 W to 176 W when the power shufflers were activated. Run 1 was conducted in full sun with the cloth covering approximately 80 % of the width of the module, whereas in runs 2 and 3 the cloth covered the full width, causing a more severe current limitation despite the shaded area being smaller. Runs 2 and 3 were also conducted later in the day under heavier cloud, so the results are not directly comparable between runs.

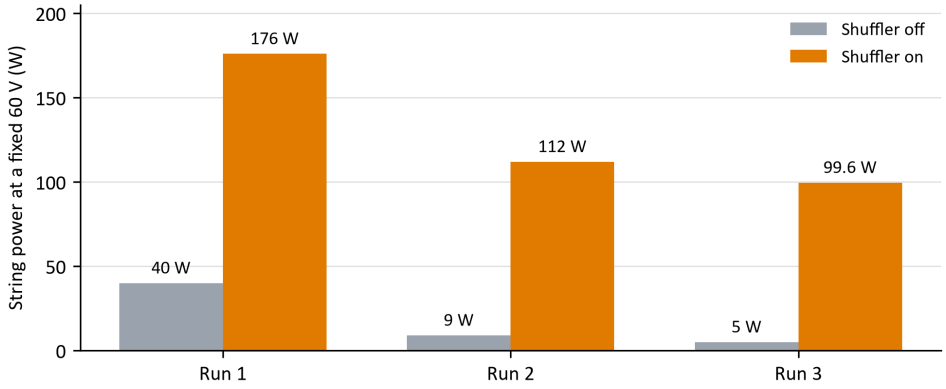


Figure 14: String output power comparison.

Figure 15 shows the transient behaviour during run 3, where passing cloud caused the string output to fall from approximately 95 W to 35 W before the power shuffler was eventually disabled. Throughout this period the module voltages remained within 0.9 V of each other, demonstrating that the power shuffler held the module voltages through the irradiance transients without any control action. The inductor current telemetry (bottom) shows the shuffled current falling as the irradiance dropped. The two legs of shuffler 1 did not share the shuffled current: ILA1 read approximately –2 A with considerable noise whilst ILB1 read close to zero. One of these channels is therefore reading incorrectly; the cause could not be established during the project.

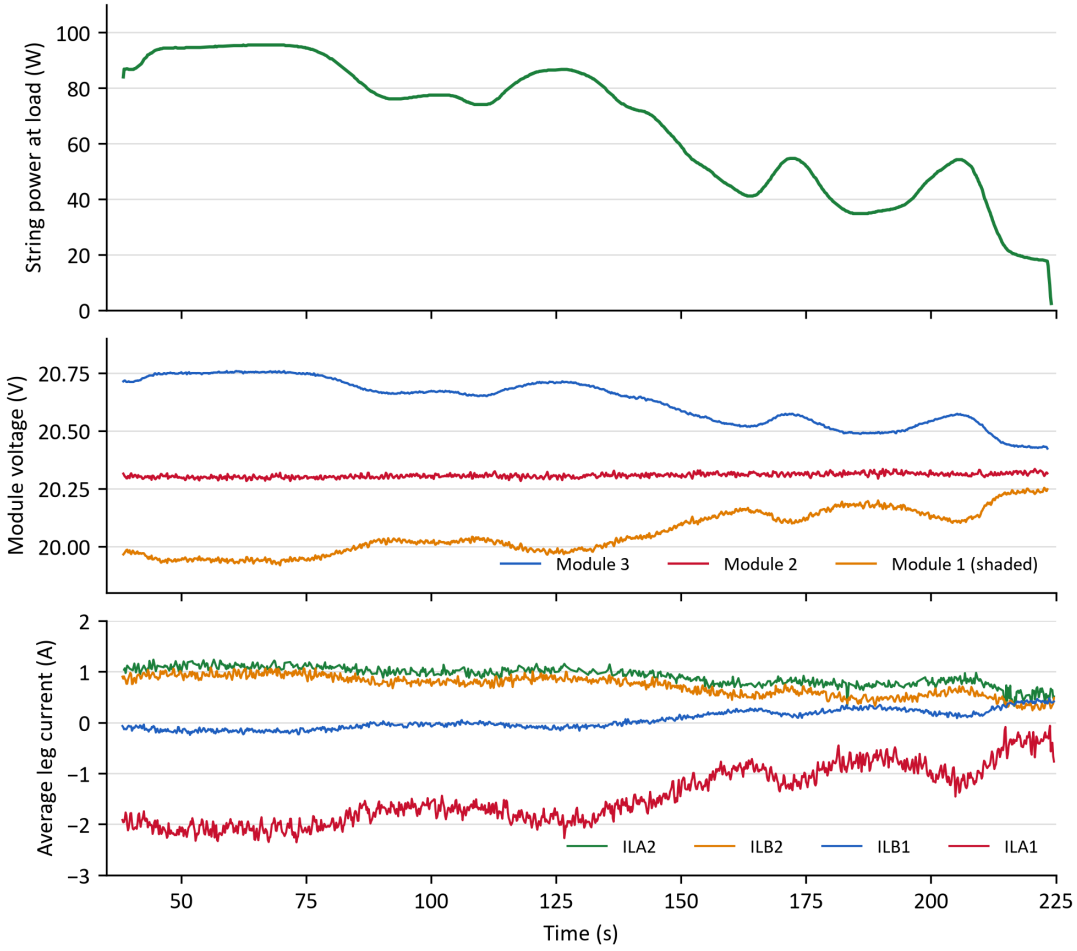


Figure 15: PV module test (run 3): string power (top), module voltages (middle) and average leg currents (bottom).

All comparisons were made at a fixed string voltage with no MPPT. As such, the gain over a tracked string was not measured and is left for future testing.

4.8 Discussion and limitations

This report addressed the gap identified in the literature review, where circuit-level design detail and hardware validation of parallel-connected partial-power converters for PV applications are seldom documented. The PCB demonstrated open-loop shuffling on emulated and real modules, with approximately 94 % of the input power reaching the load in both cases, and the measured inductor current ripple agreed with the analytical model. During testing, the response of the average inductor currents, module voltages and output power to changing input power was compared with the simulations, and the two were found to agree. The tests also exposed trade-offs which system-level studies do not report: a higher switching frequency reduces ripple and inductor loss, a fixed dead time distorts the module voltages, and fixed losses lower the efficiency at light load.

The main limitations of this project are as follows:

- Most tests used a laboratory power supply, with only limited testing on real PV modules, which was also affected by the weather;

- Comparisons were made at a fixed string voltage without an MPPT, so the true power gain on real PV modules could not be measured;
- The telemetry sampling rate was low, so the data logging could not capture fast transients;
- Telemetry was only available as Grafana screenshots, due to limitations of the logging platform;
- Inductor temperature was not measured directly and was only estimated by touch; and
- The top module's voltage sensor was wired incorrectly, which led to the failure of both the digital isolator and the sensor and limited testing to lower-output modules.
- The oscilloscope used during panel testing (Keysight InfiniiVision DSOX2004A) has an insufficiently fast sampling rate for the purpose of waveform transient analysis and no waveform captures were able to be obtained during this testing stage.

5 Conclusions and Recommendations

In the first semester, a validated simulation basis for the hardware was developed. A datasheet-driven PV model of a modern high-output module was built and validated, and the interleaved full-bridge buck–boost topology was selected as its inductor ripple was well below that of the non-interleaved alternatives. The ideal model recovered a peak of approximately 545 W over the bypass-diode string on a 2.1 kW four-module string. Realistic devices only slightly reduced this, whilst showing that shuffling should stop under extreme mismatch.

In the second semester, this basis was turned into hardware. The two-shuffler board was designed, built and tested, with its decoupling and snubbers sized using a parasitic-aware model. In open-loop bench tests, the board delivered 93.8 % of the power supplied by an emulated shaded string, with each channel delivering up to its own current limit rather than the stronger channels being limited by the weakest. On real PV modules, the output of a partly shaded string at a fixed 60 V rose from 40 W to 176 W, and in another run 93.6 % of the 106.4 W supplied by the panels reached the load. In addition, the tests showed that a higher switching frequency reduces ripple and inductor loss whilst a fixed dead time distorts the module voltages, and that fixed losses dominate at light load.

The following future work is recommended:

- adaptive dead time that follows the shuffled current, replacing the fixed 200 ns used on the prototype;
- variable switching frequency, lowering the frequency as the shuffled current rises so that the legs keep zero-voltage switching under heavy mismatch;
- integration with an MPPT converter, which was removed from this project's scope;
- adding a hardware overcurrent trip, restoring isolation for the top-module voltage sensor, and verifying the current telemetry against a direct measurement; and
- the design and assembly of a simpler single-shuffler board, with boards interlinked through headers or terminal blocks.

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Appendices

1 Research Project Timeline

All first-semester simulation deliverables were completed, namely the PV model, the topology comparison, and the ideal and semi-realistic power shuffler models. The first PCB schematic was deferred to the first two weeks of the mid-year break due to the inductor change and the late microcontroller interface requirements; however, this slip was absorbed by the buffer weeks in the plan. The planned and actual milestones are compared in Table 5 below and shown as a Gantt chart in Figure 16.

Table 5: Planned and actual milestones across both semesters (planned weeks from the project proposal)

Milestone	Planned	Actual	Notes
Project set-up and literature review	Sem 1, weeks 1–7	Completed	Proposal and literature review submitted.
System architecture (M2)	Sem 1, week 8	Completed	
Shuffler simulations and topology selection (M3)	Sem 1, weeks 9–12	Completed	PV model, topology comparison, ideal and semi-realistic models.
First PCB schematic and BOM	Sem 1, week 12	Summer, weeks 1–2	Deferred: inductor change and late controller interface.
PCB design, revisions and ordering	Sem 2, weeks 1–6	Sem 2, to 19 September	Five-input design dropped; two-shuffler board ordered on 19 September. The parasitic Qspice model was built over the following days to size the RC snubber and HF decoupling capacitors before they were ordered.
Board assembly and bench testing	Sem 2, weeks 7–10	4 to 16 October	Bring-up fixes, then open-loop tests at 100 kHz to 400 kHz
Live PV string testing	Sem 2, weeks 11–13	20 October	Three 80 W panels, below 120 V; one session, limited by weather
Final report and presentation (M9, M10)	Sem 2, week 13	Week 13	

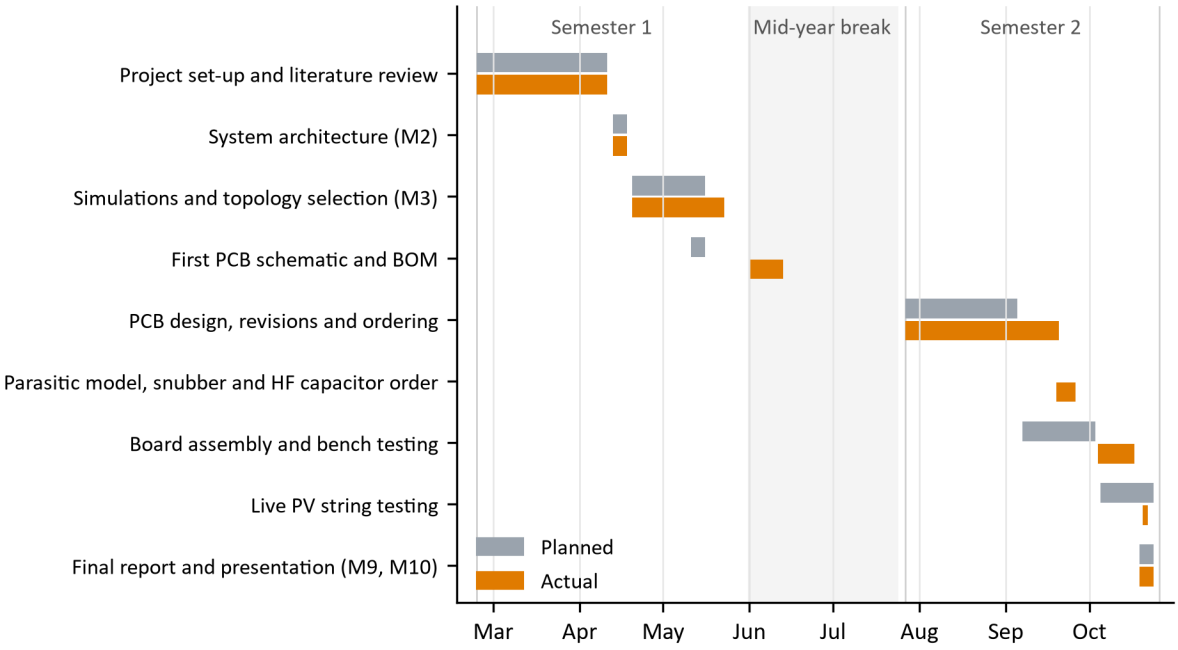


Figure 16: Gantt chart of the planned (grey) and actual (orange) project schedule across both semesters.

2 First-Semester Simulation Schematics

These schematics support Sections 3.3, 3.4 and 3.6, and were moved out of the main body to save space.

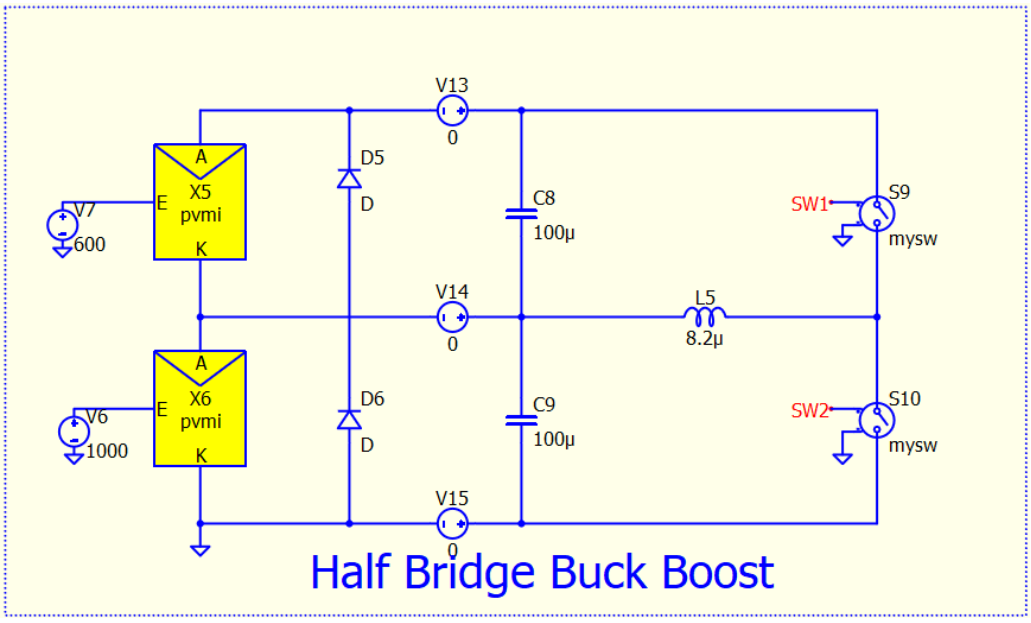


Figure 17: Half-bridge buck–boost candidate topology in Qspice.

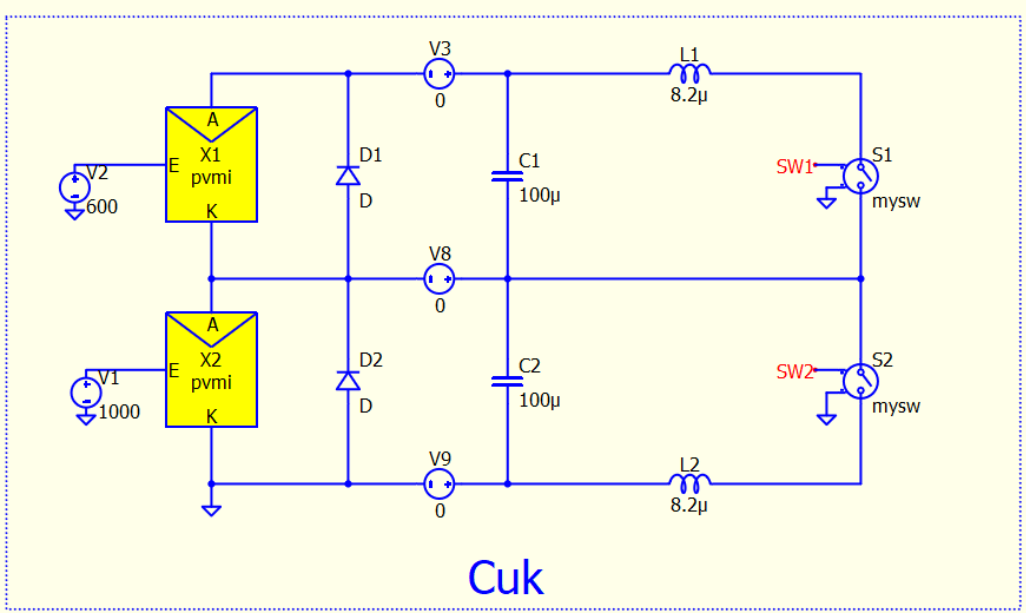


Figure 18: Ćuk converter candidate topology in Qspice.

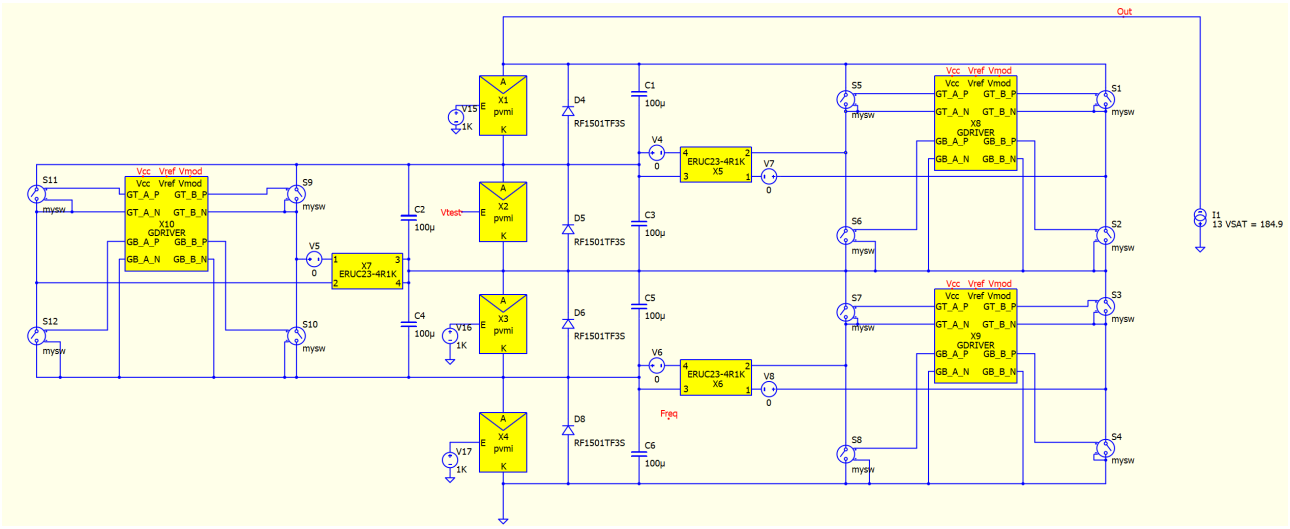


Figure 19: Ideal interleaved full-bridge buck–boost shuffler in a four-module string (Qspice)

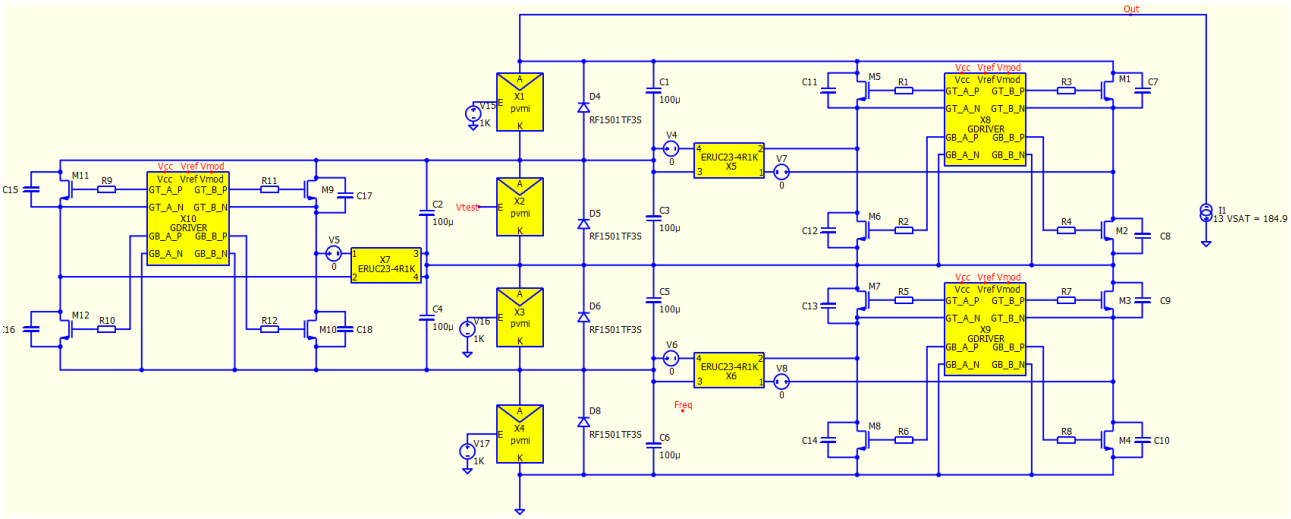


Figure 20: Semi-realistic shuffler model in Qspice

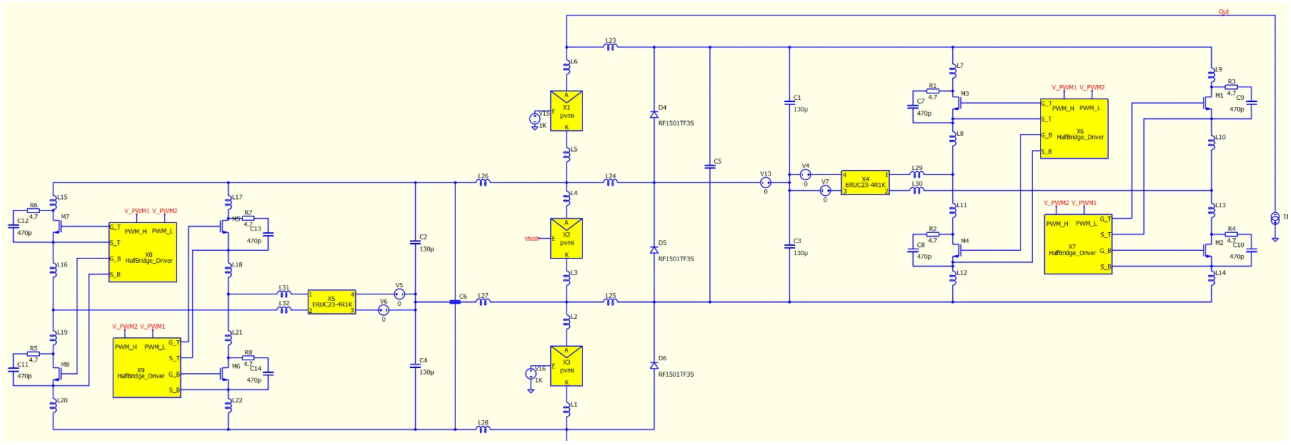


Figure 21: Parasitic-aware power shuffler model in Qspice, used to size the HF decoupling capacitors and RC snubbers (Section 3.6).

3 Test Panel Specifications

The real-panel tests used three KickAss 12 V 80 W Ultra Slim fixed glass solar panels, part number KAUSSP80. The manufacturer's ratings are a maximum power of 80 W, an open-circuit voltage of 25.07 V and a short-circuit current of 4.35 A, with monocrystalline PERC cells and dimensions of 635 × 795 × 5.9 mm.

Product page: https://kickassproducts.com.au/products/kickass-80w-ultra-slim-fixed-glass-solar-panel

User manual: https://cdn.shopify.com/s/files/1/0845/2341/7879/files/KAUSSP-USERMANUAL.pdf

4 Bill of Materials

Table 6 lists the bill of materials used for the component order placed on 19 September. The quantity listed is the number of positions on the assembled board; the cost of each line is the amount spent in the order, which included spares to cover assembly losses and rework. The 100 nF capacitors in this order are 50 V parts; once the parasitic model was completed (Section 4.5), the 24 HF decoupling positions (C61 to C84) and the eight 470 pF snubber capacitors were ordered separately as 250 V parts. The 22 Ω turn-on resistors were not ordered. The gross cost of this order was AUD 413.23.

Table 6: Bill of materials: quantity of each part required by the board and the cost of that line in the component order

Item	Part number	Designators	Quantity	Cost (AUD)
Isolated gate driver	1ED3140MU12F	U4A_B1, U4A_B2, U4A_T1, U4A_T2, U4B_B1, U4B_B2, U4B_T1, U4B_T2	8	19.44
Power MOSFET, 100 V, 45 A	STD45N10F7	Q1–Q8	8	37.44
Coupled inductor, 4.1 µH	ERUC23-4R1K	L1, L2	2	29.46
Bulk capacitor, 130 µF 80 V polymer	B40910A0137M000	C1–C4	4	20.28
10 µF ceramic, 1210	12105C106KAT2A	C6, C7, C9, C10, C12, C13, C19, C20, C25–C29, C32, C33, C35, C36, C42, C43, C48, C49, C57–C59	24	26.05
10 µF ceramic, 2220 (isolated supply)	C5750X7S2A106K230KB	C30, C31, C55, C56	4	20.82
100 nF ceramic, 0805, 50 V	KGM21NR71H104KM	C11, C14–C18, C21–C24, C34, C37–C41, C44–C47, C50–C54, C61–C84	49	19.35
Hall-effect current sensor	TMCS1126B4AQDVGRQ1	U4, U5, U8, U9	4	44.10
Power monitor	INA239AQDGSRQ1	U10, U11, U12	3	36.60
Six-channel digital isolator	ISO7762QDBQRQ1	U13	1	11.62
3.3 V LDO regulator	TPS7A4533KTTR	U2, U6	2	15.04
Wide-input buck module, 12 V	173951275 (Würth)	U14, U15	2	22.50
Bootstrap diode, 100 V Schottky	VSSA210-E3/61T	D3, D5, D8, D10	4	5.70
LED, red	LSM0805412V	D1	1	1.08
LED, green	LSM0805452V	D4	1	0.82
LED, blue	LSM0805463V	D2	1	1.32
10 Ω gate turn-on resistor, 0805	SG73S2ATTD10R0F	R12–R14, R18–R20, R35–R37, R41–R43	12	3.50
4.7 Ω turn-off and snubber resistor, 0805	SG73P2ATTD4R70F	R15, R16, R21, R22, R38–R40, R44–R52	16	4.50

Table 6 continued

Item	Part number	Designators	Quantity	Cost (AUD)
1 kΩ resistor, 0805	ERA-6AEB102V	R8, R10, R26, R31, R33	5	1.20
1.5 kΩ resistor, 0805	ERA-6AEB152V	R9, R11, R32, R34	4	0.90
680 Ω resistor, 0805	RK73H2ATTD6800F	R23, R25	2	–
100 Ω resistor, 0805	RK73H2ATTD1000F	R24	1	0.15
SPDT slide switch	L101011MS02Q	SW1	1	12.26
DP3T slide switch	L203011MS02Q	S2	1	22.14
XT60PW PV connector	Amass XT60PW	XT60_1–XT60_4	4	27.65
Test point, through-hole	Keystone 5014	TP1–TP4, TP11–TP14	8	–
Test point, SMD	Keystone 5019	TP5–TP10	6	–
Development board	NUCLEO-G474RE	U1	1	29.31
Total			**179**	**413.23**